

Day 89-C: Large-Particle Artifact Contract

VSEPR-SIM Work Order Authority

Day 89

1 Authority

Day 89-C extends the renderer-neutral `vsepr.dual_backend_3d.v1` package without transferring scientific ownership to any renderer. The package remains `manifest.json`, `scenes.csv`, and `particles.csv`; report and presentation files are sidecars.

2 Required invariants

The manifest now declares total particle count and measured generation milliseconds. Total particle count must equal both the sum of scene particle counts and the number of particle data rows. Scene identifiers represented by particle rows must equal the scene table identifiers. State values remain bounded in $[0, 1]$ and numeric serialization remains locale independent.

Invariant	Machine check
Scene count	manifest equals scene rows
Particle count	manifest equals CSV rows
Scene totals	sum equals particle rows
Stress scale	total and largest scene exceed 1,000
State range	every value in $[0, 1]$

3 Compatibility

The default package remains four scenes and 111 rows. Stress mode appends a fifth scene and uses the same columns, allowing OpenGL, BGFX, native CPU/GDI, R, XLSX, and static HTML consumers to read either package without schema branching.

4 Validation interface

The validator is invoked as:

```
python tests/test_dual_backend_3d_artifacts.py  
    out/day89_recheck --stress
```

Stress mode additionally requires more than 1,000 total rows and a scene above 1,000 particles. Report checks continue to ensure PNG, PDF, and XLSX sidecars are genuine nonempty outputs.

5 Evidence

The accepted Day 89-B package contains five scenes and 1,442 particle rows. Its manifest reports the same count and a positive generation duration. The shared contract documentation has been updated so consumers can rely on these fields rather than recomputing scale ad hoc.

6 Exit decision

Day 89-C is **COMPLETE**. General and stress validators accept packages produced by the same executable, and no renderer-specific data is added to scientific rows. Day 89-D/E may expand OpenGL and BGFX consumption while preserving this contract. W–Z remain reserved.